

Estimation of Quantum Fisher Information by variational quantum circuits on a quantum computer

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In collaboration with the Italian Space Agency (ASI)

Project: *Quantum Computers for Space Exploration*



Applications

Question: How much information about an unknown parameter can we actually access?

Quantum magnetometry

Estimate a magnetic field with an interacting spin system.

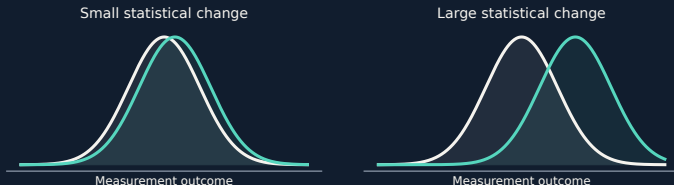
Exoplanet detection

Resolve a faint source with spatial-mode measurements.

Common workflow combining numerical simulation, quantum circuits and experimental data.

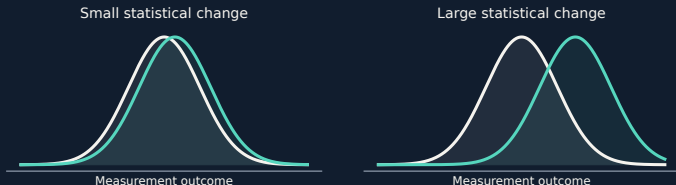
Classical Fisher Information

An unknown parameter determines the distribution of possible measurement outcomes.



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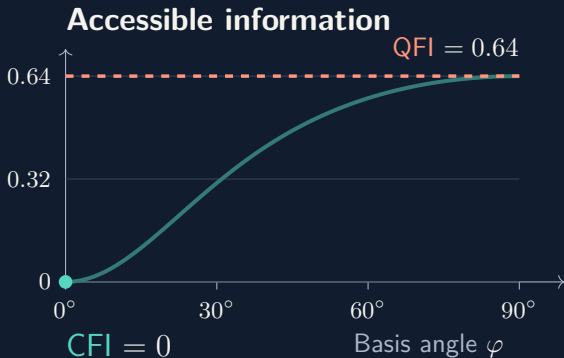
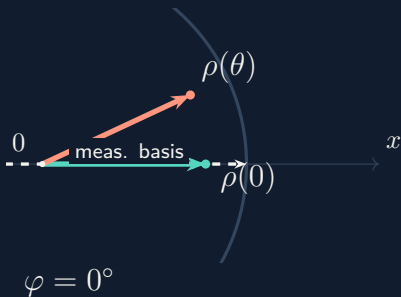
Fisher Information

Sensitivity of an observed probability distribution to changes in an unknown parameter

How much information **a specific, fixed measurement** reveals about an unknown parameter governing a probability distribution.

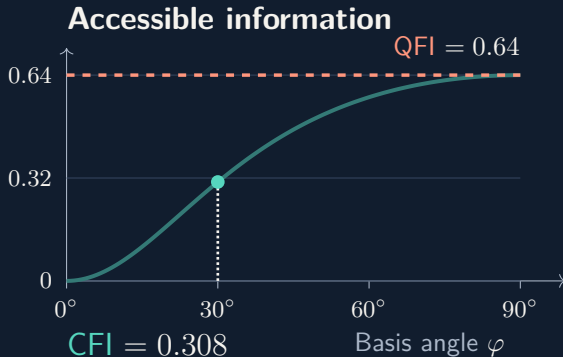
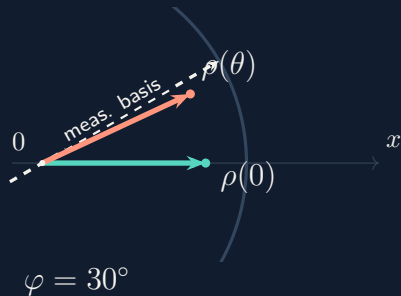
Fisher Information and Measurements

Two states at two parameter values.



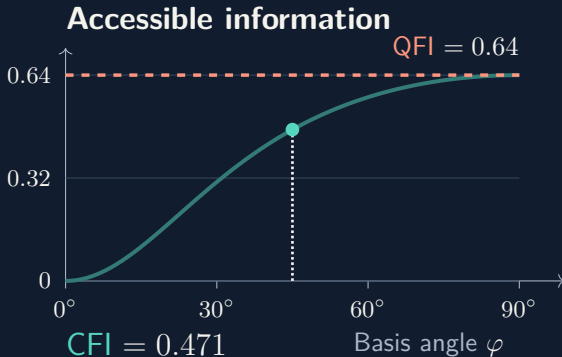
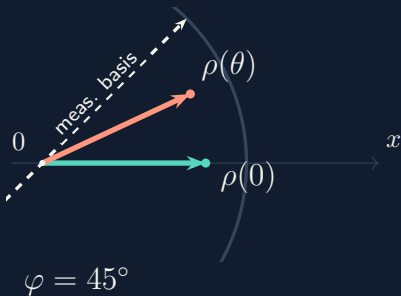
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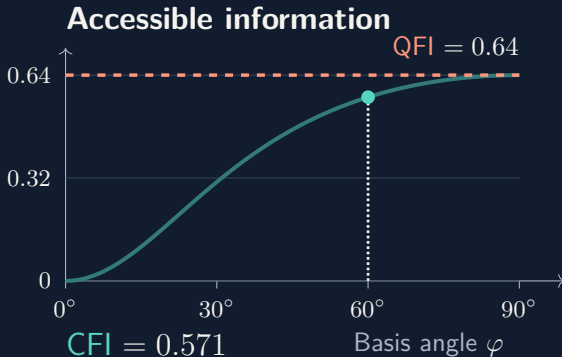
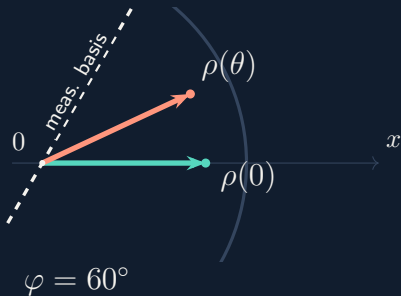
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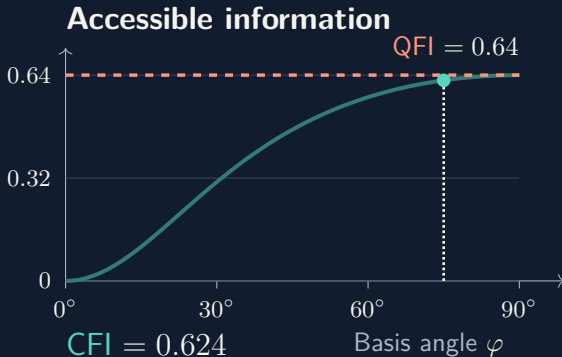
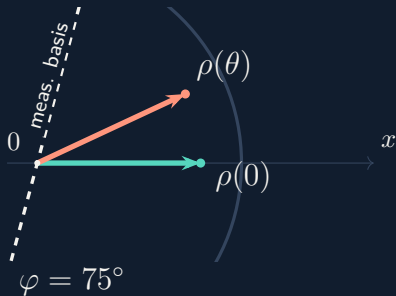
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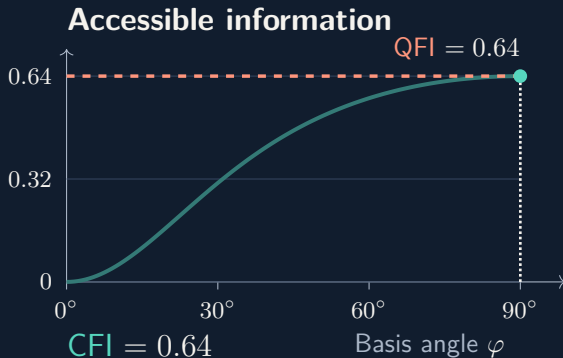
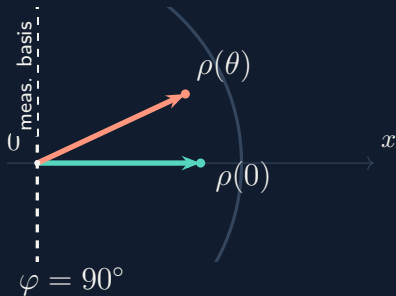
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Fisher Information and Measurements

Two states at two parameter values.



QFI is intrinsic to the quantum state
and it is **independent of the measurement basis**

The state has not become more informative, we have become better at reading it

Mixed states

WHAT Statistical mixtures of quantum states, described by a density matrix ρ

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WHY Real sensors are noisy and only partly accessible
Exact QFI becomes costly as the system grows

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WHAT Statistical mixtures of quantum states, described by a density matrix ρ

WHY Real sensors are noisy and only partly accessible
Exact QFI becomes costly as the system grows

HOW Use truncated spectral decomposition

$$\rho_m = \sum_{i=1}^m \lambda_i |\psi_i\rangle \langle \psi_i|$$

The m largest eigenvalues
and their eigenvectors.

VQSE — Variational Quantum State Eigensolver

Learns the leading eigenvalues and a circuit that prepares their eigenvectors.

TQFI — Truncated Quantum Fisher Information

Uses the retained components to construct fidelity-based QFI bounds.

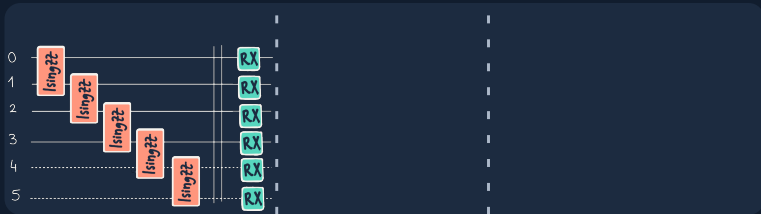
Numerical and quantum-circuit workflow

NUMERICAL SIMULATION

Preparation

Hamiltonian and
time evolution

QUANTUM CIRCUIT



Numerical and quantum-circuit workflow

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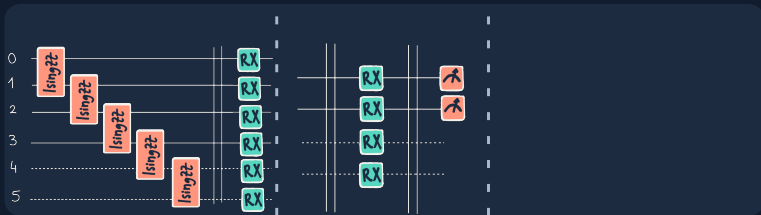
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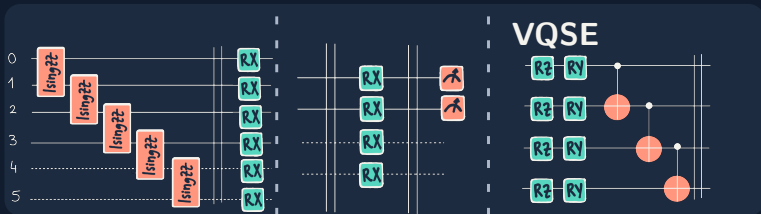
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Exact spectral
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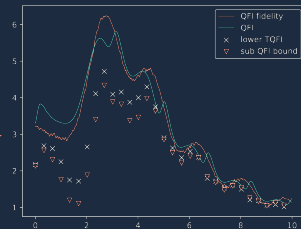
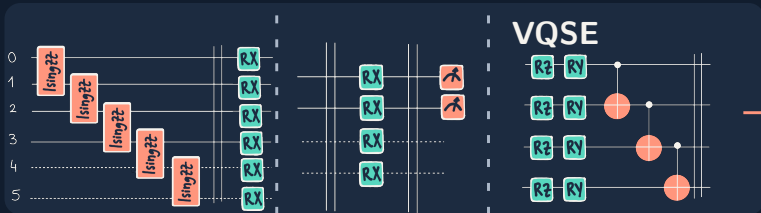
Spectrum

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Information

- 1) Exact SLD QFI
- 2) TQFI bounds

QUANTUM CIRCUIT

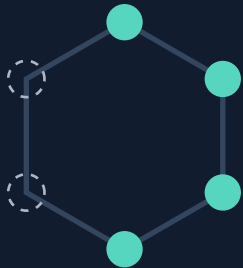


APPLICATION I

Quantum Magnetometry

A noisy quantum magnetometer

Real application: CERN, ESA's juice

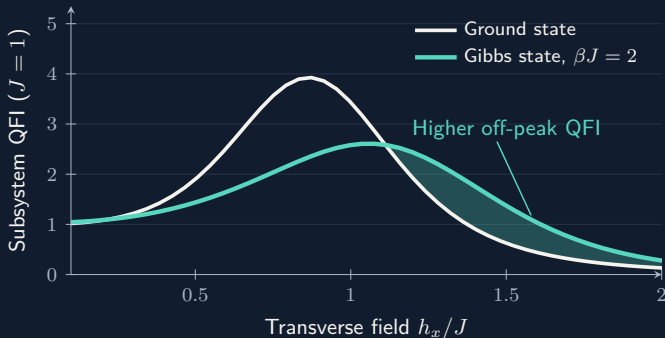


Periodic spin chain
Partial Trace: 2 discarded

Transverse-field Ising model (TFIM)

Ground state	Lowest-energy preparation
Thermal	Field-dependent populations
Depolarization	Fixed depolarizing noise

Thermal preparation broadens the sensitive range



Broader sensitivity

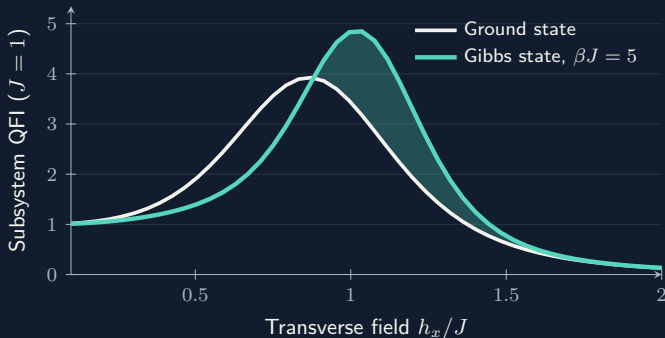
More QFI beyond the ground-state peak.

Uncertain field

Useful across a wider field range.

At $\beta = 2$, thermal preparation retains more information beyond the ground-state peak.

Cooling strengthens the sensitivity peak



Stronger peak

Higher QFI around the thermal optimum.

Known field range

Useful near the target field.

At $\beta = 5$, the thermal state exceeds the ground-state peak in the same spin subsystem.

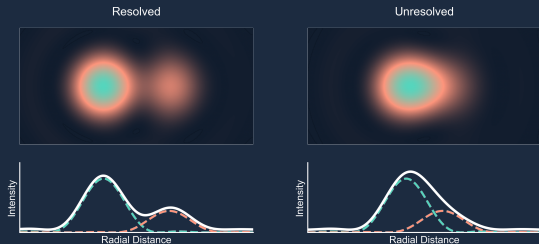
APPLICATION II

Exoplanet Detection

The Exoplanet Problem

A faint exoplanet must be resolved in the presence of a bright star.

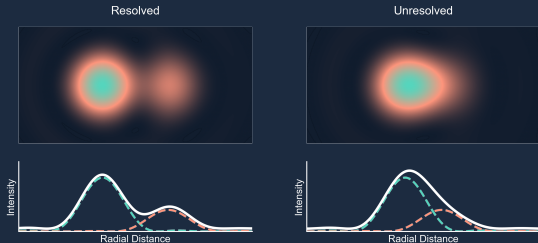
Direct imaging, Rayleigh limit



The Exoplanet Problem

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Direct imaging, Rayleigh limit



Spatial-mode sorting



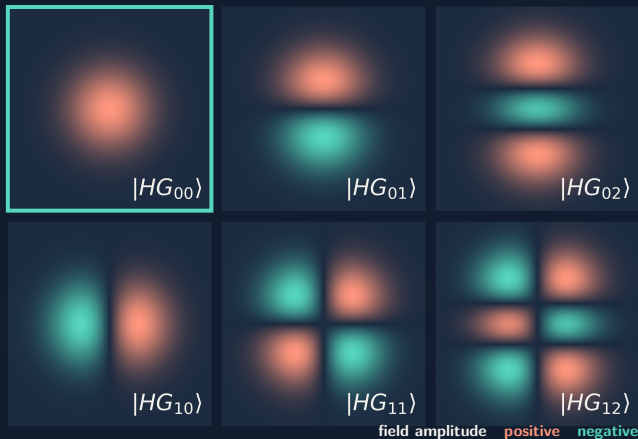
Photon counts in a spatial-mode basis.

Change the measurement basis. We used SPAtial-mode DEmultiplexing (SPADE)

SPADE, Hermite-Gauss modes and cross-talk

p_0 : **fundamental mode**

Probability of a photon in HG_{00}
dominated by the aligned star



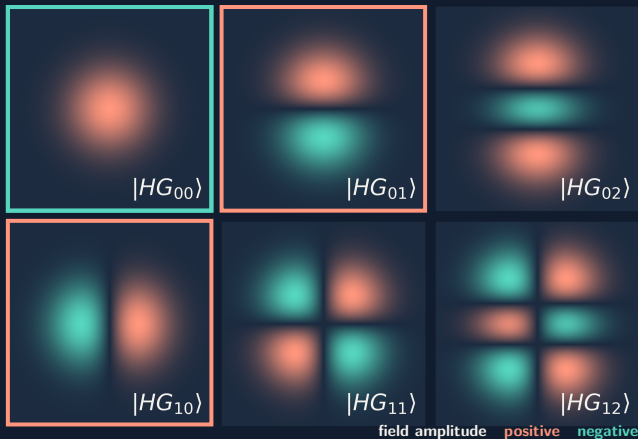
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Probability of a photon in HG_{00}
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p_1 : **first-order modes**

Probability in HG_{01} or HG_{10}
sensitive to exoplanet



SPADE, Hermite-Gauss modes and cross-talk

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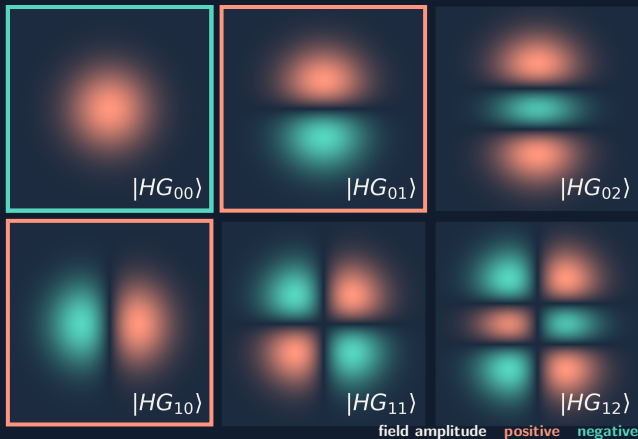
p_1 : **first-order modes**

Probability in HG_{01} or HG_{10}
sensitive to exoplanet

χ : **readout cross-talk**

Probability of reporting port 1
instead of 0, or vice versa

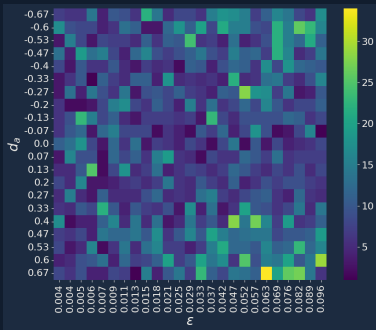
$$p_1^{\text{rep}} = \chi p_0 + (1 - \chi) p_1$$



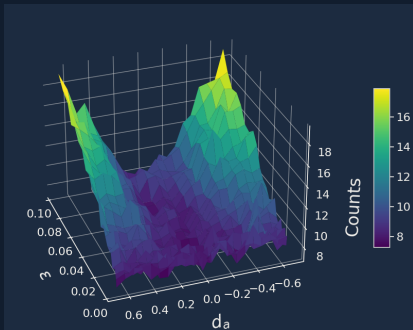
What the experiment recorded

Aggregate first-order counts, in experimental lab settings, over separation d and intensity ratio ϵ
52,500 count observations: 525 settings $\times$ 100 repetitions; 10 ms each

ONE REPETITION



MEAN OF 100



FOUNDATIONAL WORK

Fit the counts using quantum channels

Find the optimal basis

From probabilities to quantum channels

Two-mode example. Counts do not determine the whole state.

PROBABILITIES

$$\mathbf{q} = \begin{pmatrix} q_a \\ q_b \end{pmatrix}$$

Probabilities of the two mode outcomes.

DIAGONAL STATE

$$\rho_{\text{diag}} = \begin{pmatrix} q_a & 0 \\ 0 & q_b \end{pmatrix}$$

A density matrix with the same populations.

QUANTUM STATE

$$\rho = \begin{pmatrix} q_a & c \\ c^* & q_b \end{pmatrix}$$

Same populations, different coherence.

The optical source state

The state is defined before loss, measurement and detector processing.

PHYSICAL SOURCE

- **centered star**
 $|HG_{00}\rangle$
- **exoplanet**
 $|\psi_{d_a}\rangle$

ONE-PHOTON SOURCE STATE

$$\rho_S(d_a, \epsilon) = (1 - \epsilon)|HG_{00}\rangle\langle HG_{00}| + \epsilon|\psi_{d_a}\rangle\langle\psi_{d_a}|$$

$$|\psi_{d_a}\rangle = \sum_n c_n(d_a)|HG_n\rangle$$

No interference term between the two sources; the exoplanet component can still contain coherence between HG modes.

The quantum state contains modal structure that the recorded probability p_1 alone cannot reconstruct.

From the optical state to counts

Each stage describes one part of the receiver.

$$\rho_S \xrightarrow{\mathcal{E}} \rho' \xrightarrow{\text{SPADE}} \mathbf{p} \xrightarrow{R_\chi} \mathbf{p}^{\text{rep}} \xrightarrow{\text{calibration}} \mu$$

BEFORE MEASUREMENT

Loss and alignment

Mode mixing and dephasing

Act on the optical state.

From the optical state to counts

Each stage describes one part of the receiver.

$$\rho_S \xrightarrow{\mathcal{E}} \rho' \xrightarrow{\text{SPADE}} \mathbf{p} \xrightarrow{R_x} \mathbf{p}^{\text{rep}} \xrightarrow{\text{calibration}} \mu$$

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AFTER MEASUREMENT

Readout cross-talk

Calibration
Act on outcomes and counts.

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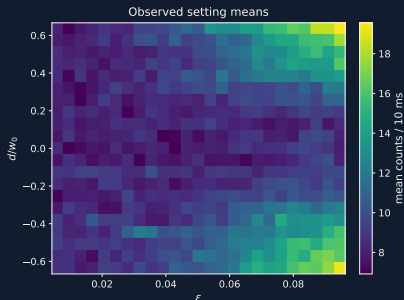
Calibration
Act on outcomes and counts.

Compare QFI before measurement with CFI after, to asses where information is lost. After the cross-talk effect, CFI/QFI drops from 90% to 57%.

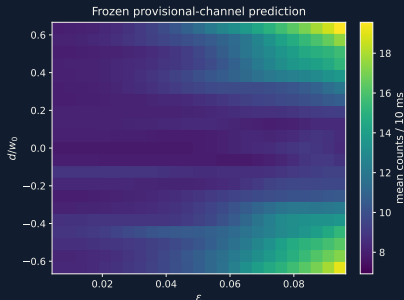
The model captures the mean count pattern

Fit to the recorded settings, with fixed optical assumptions.

Observed setting means



Modelled setting means



Agreement supports the mean-response model, not a unique reconstruction of the optical channels.

Conclusions

Simulation, preparation studies, data interpretation and a variational optical proposal.

- **Numerical and quantum-circuit sensing pipeline**

Simulated quantum sensors under noise and decoherence, estimating QFI and TQFI.

- **Preparation and noise models**

Compared ground and thermal states, depolarization and restricted access to the sensor.

- **Quantum-to-classical model of the SPADE receiver**

Specified optical channels and calibrated the observable mean-count response.

- **VQ-SPADE: a proposed optical eigensolver**

A trainable optical transformation would learn an approximate eigenbasis.

Spectral truncation and the local-QFI limit

Dominant eigencomponents bound a fidelity quantity at finite parameter displacement.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

F is root fidelity.

The retained eigenspace is state dependent; it is not a fixed detector-mode cutoff.

At regular operating points, $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$ as $\Delta \rightarrow 0$.

Finite-step bounds need not be pointwise bounds on the exact SLD QFI.

The ansatz metric and its scope

The physical sensing parameter and the trainable circuit controls define different derivatives.

$$F_{\mu\nu}^{\text{ans}} = 4 \operatorname{Re}[\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle]$$

Pure ansatz: $F^{\text{ans}} = 4g_{\text{FS}}$, the Fubini–Study metric up to convention.

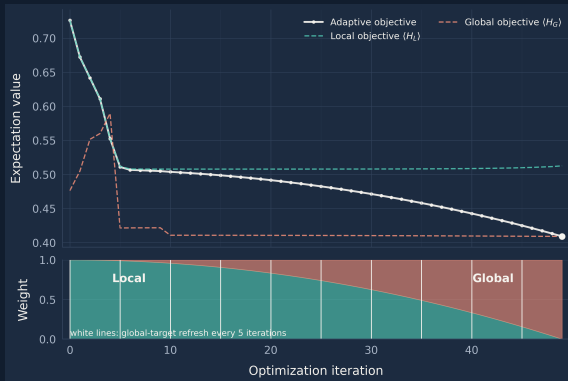
$$\min_{\Delta\alpha} \nabla C^\top \Delta\alpha + \frac{1}{2\eta} \Delta\alpha^\top (F^{\text{ans}} + \lambda I) \Delta\alpha$$

Stationarity gives the regularized natural-gradient update.

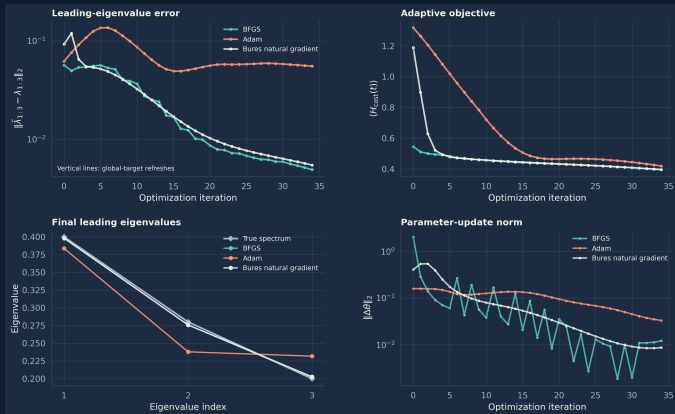
The quadratic term constrains state change, not objective curvature.

Mixed VQSE inputs require a mixed-state metric; a pure-state metric is otherwise a surrogate.

Local-to-global objective schedule



BFGS, Adam and Bures natural gradient



Thermal sensitivity and data processing

The thermal and ground-state families need not have the same field response.

$$\rho_{\beta}(h_x) = \frac{e^{-\beta H(h_x)}}{\text{Tr } e^{-\beta H(h_x)}}$$

The field changes both eigenvectors and thermal populations.

Partial trace still obeys data processing within each family:

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h_x)] \leq \mathcal{F}_Q[\rho_{\beta}(h_x)]$$

For an uncertain field, choose an operating range or prior.

These curves do not establish a universal Bayesian or minimax optimum.

The comparison changes the preparation family; it does not reverse a fixed noise channel.

Proof of Kraus-unitary freedom

A change of Kraus basis leaves the channel unchanged.

$$\widetilde{K}_a = \sum_b u_{ab} K_b, \quad U^\dagger U = I$$

$$\sum_a u_{ab} u_{ac}^* = \delta_{bc}$$

Orthonormal columns of U .

$$\sum_a \widetilde{K}_a \rho \widetilde{K}_a^\dagger = \sum_{b,c} \delta_{bc} K_b \rho K_c^\dagger = \sum_b K_b \rho K_b^\dagger$$

Isometry in the Kraus index leaves the action on every input unchanged.

A physical realization adds assumptions about the environment; the reduced map alone does not select them.

A classical response does not fix a quantum channel

Embedding a stochastic map is a model construction, not process tomography.

$$\mathbf{p} \mapsto \text{diag}(\mathbf{p}), \quad \mathbf{p}' = R\mathbf{p}$$

For a column-stochastic R , one valid quantum extension is

$$K_{ji} = \sqrt{R_{ji}}|j\rangle\langle i|, \quad \mathcal{E}_R(\rho) = \sum_{ji} R_{ji}\rho_{ii}|j\rangle\langle j|$$

This extension measures and prepares in the chosen basis, erasing coherences.

For $R = I$, both the identity channel and complete dephasing reproduce the same populations.

Different channels can agree on the measured populations; this is distinct from Kraus freedom of one channel.

The loss map and its physical checks

The input mode space is embedded in an output space containing an orthogonal failure flag.

$$\mathcal{L}_\tau(\rho) = K_t \rho K_t^\dagger + \text{Tr}[(I - K_t^\dagger K_t)\rho] |\text{fail}\rangle \langle \text{fail}|$$

For any ancillary system, Kraus form preserves positivity:

$$(\mathcal{L} \otimes \text{id})(X) = \sum_a (K_a \otimes I) X (K_a^\dagger \otimes I) \geq 0 \text{ for } X \geq 0.$$

Boundary checks: $\tau_j = 1$ transmits mode j ; $\tau_j = 0$ sends it to failure.

Normalization: transmission plus failure has unit trace.

Physical validation: mode-injection measurements determine the τ_j .

Renormalizing the transmitted branch alone changes the question to postselected performance.

Predictive calibration of the observed response

A physical mean with a Gaussian-process correction predicts held-out setting means.

$$\mu(d_a, \epsilon) = \mu_0(d_a, \epsilon) \exp[f(d_a, \epsilon) + \xi]$$

Grouped five-fold validation; all components refitted within each training fold.

Held-out structure	RMSE	Standardized SD	95% coverage
Checkerboard cells	0.606	1.004	93.52%
Distance rows	0.734	1.041	94.67%
Source-ratio columns	0.609	1.009	93.71%

The validated target is the mean of 100 repetitions; this does not identify a complete quantum channel.

From SPADE to a variational optical measurement

VQ-SPADE is a proposed optical realization of variational state eigensolving.

$$p_k(\boldsymbol{\alpha}; d_a) = \langle HG_k | V(\boldsymbol{\alpha}) \rho(d_a) V^\dagger(\boldsymbol{\alpha}) | HG_k \rangle$$

A programmable coherent mode transformation provides the trainable basis.

Resolved outputs, flux and failure events provide normalized probabilities.

Nearby states and additional overlap measurements support fidelity-based QFI bounds.

The present record motivates the acquisition design; it does not implement this protocol.

Mixed states and measurement-dependent information

Two finite-separated vectors illustrate one state family; Fisher information is evaluated locally.

$$\rho(\theta) = \frac{I + r(\cos \theta X + \sin \theta Z)}{2}, \quad r = 0.8, \quad \rho \equiv \rho(0)$$

$$M_{\pm}^{(\varphi)} = \frac{I \pm (\cos \varphi X + \sin \varphi Z)}{2}, \quad p_{\pm}(\theta) = \frac{1 \pm r \cos(\theta - \varphi)}{2}$$

$$\mathcal{F}_C(\varphi)|_{\theta=0} = \frac{r^2 \sin^2 \varphi}{1 - r^2 \cos^2 \varphi} \leq r^2 = \mathcal{F}_Q$$

At $\varphi = 0^\circ$: CFI = 0. At 45° : CFI = $8/17 \simeq 0.471$.

At $\varphi = 90^\circ$, measuring Z attains QFI = 0.64; the SLD is $L_0 = rZ$.

Both displayed vectors have radius 0.8; their 25° separation is only illustrative.

Changing the measurement changes accessible information, not the underlying quantum limit.

Kraus representations are not unique

Different operator sets can describe exactly the same quantum channel.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \quad \sum_a K_a^\dagger K_a = I$$

Example: complete dephasing of a two-mode state in the chosen basis.

Basis projectors

$$K_0 = |0\rangle\langle 0|, \quad K_1 = |1\rangle\langle 1|$$

Random phase flips

$$\widetilde{K}_0 = \frac{I}{\sqrt{2}}, \quad \widetilde{K}_1 = \frac{Z}{\sqrt{2}}$$

$$\sum_a K_a \rho K_a^\dagger = \frac{\rho + Z\rho Z}{2} = \text{diag}(\rho)$$

Kraus freedom changes the representation. Physical hypotheses concern the channel.

A physical construction: mode-dependent loss

Mode transmission $0 \leq \tau_j \leq 1$; loss is represented by an orthogonal failure state.

Choose operators with explicit physical roles

$$K_t = \sum_j \sqrt{\tau_j} |j\rangle \langle j|, \quad K_j^f = \sqrt{1 - \tau_j} |\text{fail}\rangle \langle j|$$

Transmission and failure jointly account for every input photon.

$$K_t^\dagger K_t + \sum_j (K_j^f)^\dagger K_j^f = \sum_j [\tau_j + (1 - \tau_j)] |j\rangle \langle j| = I$$

Kraus form ensures complete positivity; completeness proves trace preservation.

Channel validity can be proved mathematically; physical suitability requires calibration.

Information retained by the optical receiver

Conditional model for d_a , per surviving photon, across nine settings.

$$\mathcal{F}_Q(\rho_S) \geq \mathcal{F}_Q[\mathcal{E}(\rho_S)] \geq \mathcal{F}_C(\mathbf{p}) \geq \mathcal{F}_C(R\mathbf{p})$$

For channels and measurements independent of the parameter being estimated.

Checkpoint

Fraction of source QFI

● Resolved reference, first-order and failure outputs	99.68–100%
● Binary readout; assumed 0.35% port confusion	0.13–57.0%

Aggregation and readout confusion can strongly reduce the accessible information.

From information to precision: the quantum Cramér–Rao bound

QFI is the best local Fisher information available over all quantum measurements.

FIXED POVM

$$\text{Var}(\hat{\theta}) \geq \frac{1}{N\mathcal{F}_C(\theta; M)}$$

Assuming N independent trials and local unbiasedness.

BEST MEASUREMENT

$$\mathcal{F}_C(\theta; M) \leq \mathcal{F}_Q(\theta)$$

Maximize the CFI over all POVMs for the same state family.

QUANTUM LIMIT

$$\text{Var}(\hat{\theta}) \geq \frac{1}{N\mathcal{F}_Q(\theta)}$$

Equivalently, $\Delta\theta \geq 1/\sqrt{N\mathcal{F}_Q}$.

The QCRB is an ideal local precision limit; attaining it also requires a suitable measurement and estimator.

When modal counts become spectral information

SPADE populations are eigenvalues only under a specific basis assumption.

1. MEASURE

$$p_j = \langle HG_j | \rho | HG_j \rangle$$

SPADE returns populations in the chosen HG basis.



2. ONLY IF

$$\rho = \sum_j \lambda_j |HG_j\rangle \langle HG_j|$$

HG is an eigenbasis. Record the modes separately with known total normalization.



3. RECONSTRUCT

$$\rho_r = \sum_{j=0}^{r-1} p_j |HG_j\rangle \langle HG_j|$$

Here $p_j = \lambda_j$. For TQFI, keep the modes with the largest eigenvalues.

TQFI keeps dominant eigenvectors. They need not be HG modes.

Dephasing and readout act at different stages

Optical effects act on the state; readout cross-talk acts on probabilities.

$$\rho_S \xrightarrow{\mathcal{E}} \rho' \xrightarrow{\text{SPADE}} \mathbf{p} \xrightarrow{R_\chi} \mathbf{p}^{\text{rep}} \xrightarrow{\text{calibration}} \boldsymbol{\mu}$$

QUANTUM CHANNEL

$$\begin{pmatrix} q_a & c \\ c^* & q_b \end{pmatrix} \xrightarrow{\mathcal{D}_v} \begin{pmatrix} q_a & vc \\ vc^* & q_b \end{pmatrix}$$

Before SPADE: dephasing scales c by $0 \leq v \leq 1$ while leaving two HG-mode populations unchanged. Other optical maps describe loss, alignment, coherent mixing and modal transmission.

CLASSICAL CROSS-TALK

$$\mathbf{p}^{\text{rep}} = R_\chi \mathbf{p}$$

$$p_1^{\text{rep}} = \chi p_0 + (1 - \chi) p_1$$

After SPADE: cross-talk redistributes outcomes after coherence is inaccessible to this measurement.

A probability-only model cannot describe pre-measurement coherence or its loss.

Photon loss as an erasure channel

A photon either reaches SPADE or is transferred to the vacuum outcome.

$$\rho_S \xrightarrow{\mathcal{S}_\eta} \rho_{\text{surv}} \xrightarrow{\text{SPADE}} \mathbf{p} \xrightarrow{R_\chi} \mathbf{p}^{\text{rep}} \xrightarrow{\text{detector}} \mu$$

QUANTUM LOSS CHANNEL

$$\mathcal{S}_\eta(\rho_S) = \eta \rho_S + (1 - \eta) |\text{vac}\rangle \langle \text{vac}|$$

η : the photon survives

$1 - \eta$: the photon is lost

The vacuum branch keeps total probability equal to one.

WHAT THE COUNTS CONSTRAIN

$$\mu = B + T g(d, \epsilon) p_1^{\text{rep}}$$

$$T = N_{\text{in}} \eta$$

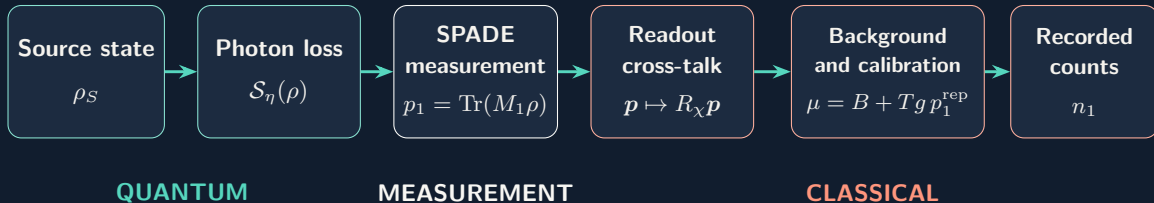
T : fitted effective throughput

B : fitted effective background

Dark counts belong to the detector layer and may contribute to B ; they are not the optical loss channel.

From the source state to recorded counts

The model separates optical evolution, measurement and detector effects.



Each stage acts on a different object: quantum state $\rightarrow$ probabilities $\rightarrow$ count statistics.